

Sub-critical crack growth in some phosphate glasses

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Available online 14 November 2007

Abstract

The effect of alumina on the sub-critical crack growth is investigated on a set of calcium aluminophosphate glasses of molar composition $50\% \text{P}_2\text{O}_5 - (50 - x)\% \text{CaO} - x\% \text{Al}_2\text{O}_3$ ($0\% \leq x \leq 10\%$). The crack propagation is operated using the double-cleavage-drilled-compression (DCDC) method. In this test, the sample is loaded in compression with a mechanical testing machine in an environmentally controlled chamber (both temperature and water vapor pressure). The crack velocity is plotted as a function of stress intensity factor. Regions I and II are characterized by a dependence of crack velocity on the amount of alumina content in the glass. When the alumina content increases, glass shows, as expected, a trend less sensible to the stress corrosion correlated to the slope of the crack growth curves, but in the same time, these curves are shifted toward higher crack velocity values. For different compositions, an unexpected behavior is also observed with temperature and water vapor pressure. We assume the formation of a 'gel-like product' on fresh fracture surfaces which modifies the reactivity at the crack tip.

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PACS: 61.43.Fs; 62.20.mm; 62.20.mt; 81.40.Np

Keywords: Phosphate glasses; Mechanical properties; Crack growth; Fatigue; Fracture

1. Introduction

It is generally accepted that water or water vapor attacks the region around the crack tips of surface flaws, weakening the structure in the vicinity of the crack tip. This allows cracks to grow at stress intensities below the critical value (K_{IC}). This stress corrosion cracking phenomenon is quite complex due to local chemical reactions that occur and to diffusion mechanisms. The fracture process itself may activate the surface for reactions with water vapor. Charles has the first described this phenomenon on soda-lime silicate glasses under water vapor [1]. Additional works have been developed later by Wiederhorn [2]. More recently, phosphate laser glasses have been studied [3–5]. Crack growth has been assumed to occur by stress-enhanced chemical reaction at the crack tip ($\text{P-O-P} + \text{H}_2\text{O} \Rightarrow 2(\text{P-OH})$ [4]). Excepted in some cases as capillary condensation phenomenon [6], crack velocity is enhanced by

increasing water vapor pressure or increasing temperature as described by the Wiederhorn's model.

Phosphate glasses offer very interesting properties for specific applications as matrix for elements giving rise laser effects [5]. However, they have lower transition temperature, fracture toughness and chemical durability than silicates glasses. Different studies were realized in order to improve this last property [7]. For example, the addition of Al_2O_3 to a sodium phosphate glass has significant effects on physical and chemical properties, including the improvement of the aqueous durability [8,9]. Because the application of a mechanical stress, even weak, can modified the global resistance of the glass by introduction of the stress corrosion cracking phenomenon, knowledge of this phenomenon is of interest for a good description and lifetime estimate of the glass.

We propose to study the influence of water vapor pressure and temperature on sub-critical crack growth of some calcium aluminophosphate glasses. The analysis is based on the reactivity of glasses containing various amount of alumina. The crack is propagated using the

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double-cleavage-drilled-compression method which was first developed by Janssen [10]. This technique offers different advantages as auto-cracking at the top and the bottom of the hole when the sample is under compressive load and an easier monitoring of the crack growth as the crack velocity decreases in time.

2. Experimental procedures

2.1. Samples preparation

The glasses used in this study are calcium aluminophosphate glasses. Their molar composition corresponds to: 50% P_2O_5 –(50 – x)% CaO – x % Al_2O_3 . Glasses without alumina are labeled CP, glasses containing x % alumina, CAP x %. These glasses were prepared using calcium dihydrogenophosphate and aluminium orthophosphate powders. Calculated amounts of these chemical compounds were thoroughly mixed and then melted in a platinum crucible at about 1250 °C. After melting for about 5 h, the glasses were poured in hot metallic mould, annealed at the appropriate temperature and then cooled slowly (30 °C/h) down to ambient temperature. Because water contents in glasses can influence slow crack growth [4], infrared spectra measurements, using absorption coefficient at 3000 cm^{-1} (120 l/mol/cm), have been done in order to control that water amount is slightly the same in all studied glasses. It is estimated about 180 ppmw. Glasses composition was analyzed by EDS and is given in Table 1.

The glass samples were cut into rectangular pieces (Fig. 1) 47.0 mm length, 4.7 mm wide and 4.1 mm thick. A 1.25 mm hole was core-drilled at the center of the 47.0 mm × 4.7 mm faces of the specimen. Samples were stored in desiccators prior to test.

In the range $0\% \leq x \leq 10\%$, the O/P ratio varies from 3 to 3.13 corresponding to a change from a metaphosphate (O/P = 3) to a pyrophosphate (O/P = 3.5) composition.

Elastic modulus (E) and Hardness (H_v) were obtained by nano-indentation. The fracture toughness (K_{IC}) was calculated using cracks initiated by a Vickers indenter and 0.1 to 0.3 kg loads. Each measurement was achieved in air at room temperature just after polishing of the surfaces.

2.2. Experimental setup

An environmentally controlled chamber (Fig. 2) permits to control temperature from ambient (± 1.5 °C) to 450 °C (± 2 °C) and water vapor pressure from 0.5 Pa to 2200 Pa

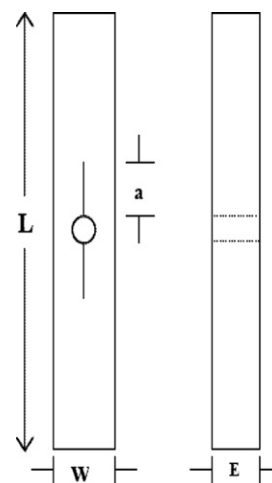


Fig. 1. Geometry for double-cleavage-drilled-compression (DCDC) specimen.

($\pm 2\%$). Under humid atmosphere, the desired water vapor pressure is provided by bubbling nitrogen gas through a water bath kept at constant temperature. The temperature is reached by heating the gas flow (N_2 + water vapor pressure) at the desired temperature prior introduction into the chamber, associated to a heating element winded around the sample-carrier. The temperature is controlled with a K thermocouple located in the vicinity of the sample.

The sample receives the stress through a ball-bearing to ensure that the load was distributed evenly across the cross-section of the sample, as opposed to the point or line loading possible if the end faces of the sample are not perfectly flat and aligned with the compression grips.

Before sample loading, temperature and water vapor conditions are achieved in the apparatus in order to avoid dilatation effect of the metallic pieces. Compression load is applied with a 5500 Instron model. The experiment consists first in slowly bringing the Instron crosshead to the contact of the sample. Then the stress is increased by lowering the crosshead at a constant rate of 0.1 mm/min until the appearance of the crack at the bottom and the top of the hole after which the stress is kept constant at the crack initiation value.

The crack front is recorded with a camera for crack velocities from 10^{-6} m/s to 10^{-3} m/s. For velocities from 10^{-9} m/s to 10^{-6} m/s, the cracked sample is projected on a screen using a lens with a 20× magnification and the crack length is measured using a cathetometer.

Stress intensities factor K_I ($MPa\ m^{1/2}$) were calculated from the applied stress σ (MPa), measured crack length from the hole's edge (see Fig 1), a (m), sample wide, w (m) and hole radius, R (m) using the relationship developed by He et al. [11]

$$K_I = \frac{\sigma \sqrt{\pi R}}{\frac{w}{2R} + (0.235 \frac{w}{2R} - 0.259) \frac{a}{R}} \quad (1)$$

Eq. (1) is useful for following ratios: $w/2R \leq a/R \leq 15$, $2 \leq w/2R \leq 4$ and $2L/w = 20$. Crack velocity was calcu-

Table 1
Glasses composition in molar percent ($\pm 0.5\%$)

	CaO	P_2O_5	Al_2O_3	O/P
CP	49.5	50.5	0	3
CAP 2%	46.9	50.7	2.4	3.03
CAP 4%	44.7	51.3	4	3.05
CAP 10%	37.7	52.6	9.7	3.13

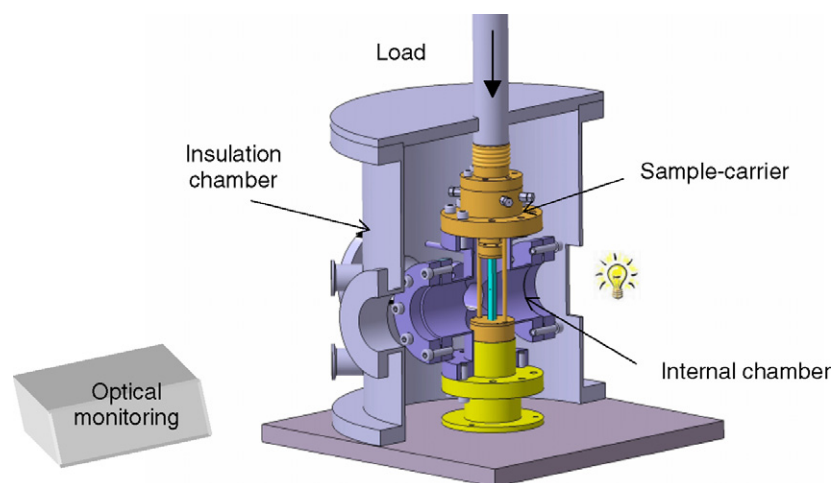


Fig. 2. Schematic of test apparatus. Section is along the optical monitoring line.

lated by a simple finite difference between successive data points, $v = \Delta a / \Delta t$, where Δa is the incremental crack length for the incremental time Δt .

Experimental uncertainties of crack rates and stress intensity factor were 5% and 6%, respectively. Experiment durations were all relatively short between 4 and 6 h.

3. Results

Glass transition temperature (T_g), Young's modulus, hardness and K_{IC} of all glasses are summarized in Table 2. With increasing Al_2O_3 content in glass, all these physical properties increase. For mechanical properties, the increase is not linear assuming that an efficiency threshold in the alumina content exists.

Stress corrosion data were analyzed plotting crack velocity versus stress intensity factor. These curves show three distinct regions, I, II and III already observed in many glasses [2–6]. Region I refers to conditions where crack growth is reaction-rate limited and a linear relation between $\log(v)$ and K_I can be found. Crack growth is limited only by the reaction-rate between H_2O and the P–O–P or P–O–M bonds located at the crack tip. In region II, the crack velocity is no longer reaction-rate limited but becomes limited by mass-transport (diffusion) of H_2O to the crack tip; the measured crack velocity in region II remains nearly constant with increasing K_I . Finally, in region III, the crack velocity becomes independent of the chemical environment. Region III typically occurs at K_I values approaching the fracture toughness limit and is evi-

denced by an abrupt increase in crack velocity. Because of the high crack velocity, this region is not always observed depending on the experiments parameters.

3.1. Dependence on water vapor pressure

The effect of water vapor pressure at ambient temperature on slow crack growth was studied for all compositions. Behaviors of CAP 4% and CP are plotted in Fig. 3. All the three regions described earlier are observed in CAP 4% at a water vapor pressure of 0.5 Pa.

Curves are related to water vapor pressure. With increasing water vapor pressure, curves are shifted to higher crack rates. For 0.5 Pa and 1320 Pa water vapor pressure, the increase is more pronounced for the CAP 4% (10^3) than for the CP (about 10). In Fig. 3(a), the crossover between region I and region II is shifted to lower K_I when water vapor pressure increases. Further increase of water content up to 2110 Pa has a significant effect for the CP.

3.2. Dependence on alumina content

Comparison of fracture behavior of all glasses in two different water vapor pressures is shown in Fig. 4. It can be seen that composition has a marked effect on the rate of crack growth. At constant stress intensities factor (K_I), the larger alumina content glass has faster crack velocities in both regions I and II. The crack velocity shift is also emphasized for a high water vapor pressure.

3.3. Dependence on temperature

The effect of temperature exhibits an interesting trend (Fig. 5). For CP and CAP 10%, whatever the water vapour pressure, curves are approximately parallel but shifted to upper K_I increasing temperature from ambient to 120 °C. For CP, when water vapor pressure is 0.5 Pa (Fig. 5(a)), the shift is not so obvious comparatively to 1320 Pa

Table 2
Some physical properties of metaphosphate glasses

	T_g (°C)	E (GPa)	H_v (GPa)	K_{IC} (MPa.m ^{1/2})
CP	516 ± 1	45 ± 5	4.2 ± 0.4	0.51 ± 0.05
CAP 2%	539 ± 3	45 ± 3	5.0 ± 0.5	0.53 ± 0.05
CAP 4%	544 ± 3	72 ± 4	7.0 ± 0.3	0.79 ± 0.03
CAP 10%	535 ± 3	84 ± 4	7.2 ± 0.2	0.81 ± 0.03

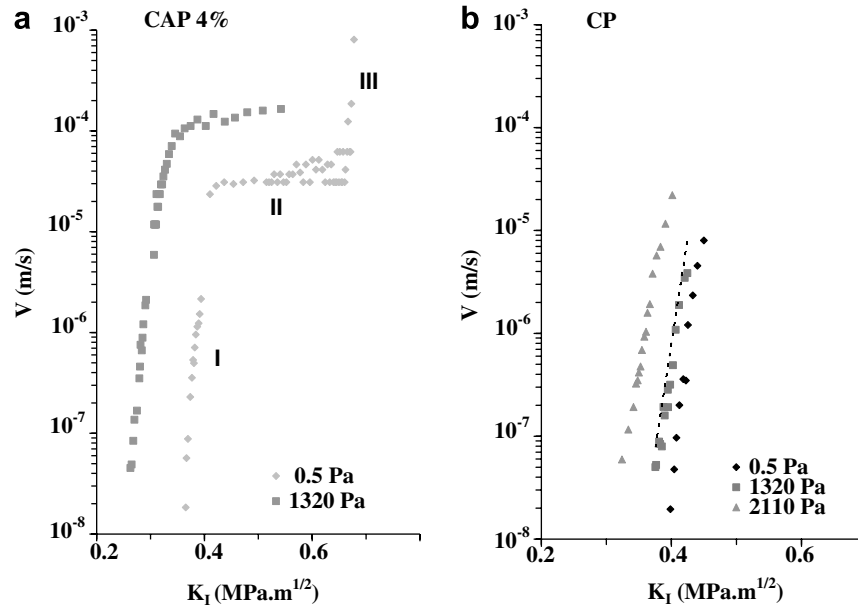


Fig. 3. Effect of water vapor pressure at ambient temperature for (a) CAP 4% and (b) CP, dash line is the predicted curve for 2110 Pa using Eq. (2).

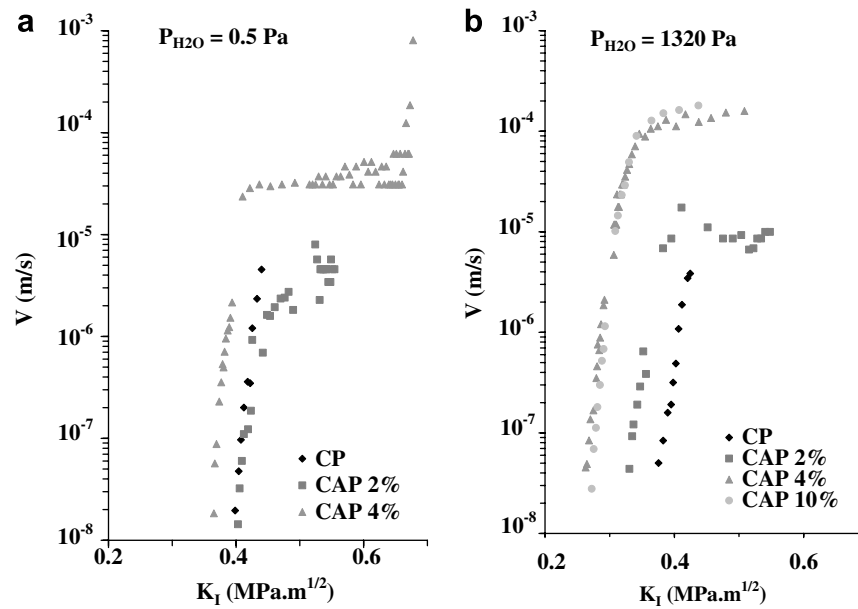


Fig. 4. Effect of alumina content at ambient temperature and under various water vapor pressures: (a) 0.5 Pa and (b) 1320 Pa.

(Fig. 5(b)). It is noteworthy that the shift is more pronounced for CP than for CAP 10% at a water vapour pressure of 1320 Pa (Fig. 5(b) and (c)). On CP glass at 1320 Pa, when the temperature is increased to 250 °C, the curve is contrarily shifted to lower K_I . At 120 °C a plateau like region is also observed for a crack rate of 3.10^{-7} m/s. Plateaus, at low crack rate, have already been observed by Gehrke [12] and Michalske [13] in some borosilicate glass. Experimental conditions were acidic solution and room temperature. They assume a corrosion phenomenon located at the crack tip area and tensile stresses in the leached layer.

Associated to all experiments achieved at 120 °C, a new phenomenon is observed: appearance of droplets along the crack line when the sample is unloaded. Thus when crack arrests and closes (right photo side in Fig. 6). Temperature and water vapor pressure values are still those of the experiment. A non-wetting liquid phase seems to be squeezed out of the crack. Droplets are then located on the crack track. Because of the poor contrast, it is not possible to estimate the total volume of droplets. In Fig. 7, the micrograph of the sample cooled to room temperature, displays chemical residues coming from previously observed drop-

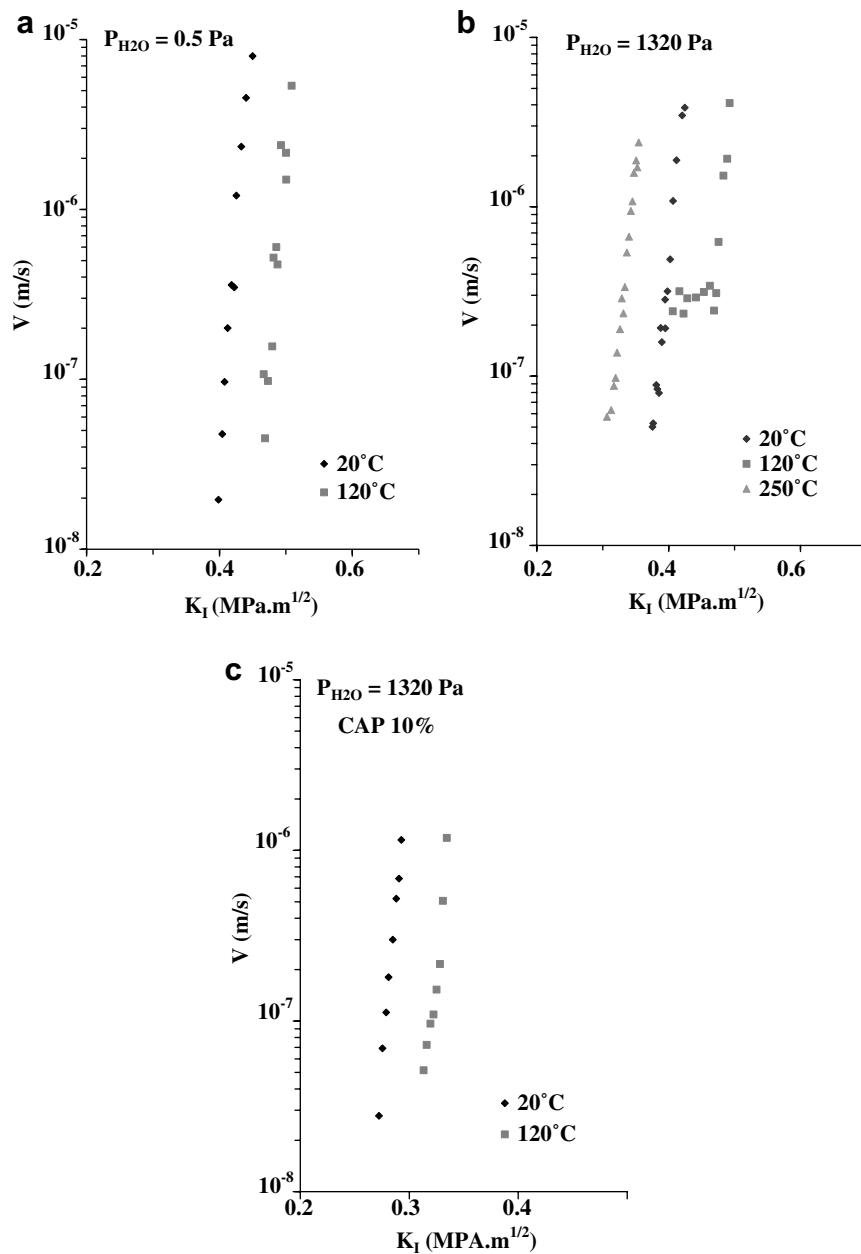


Fig. 5. Dependence on temperature for CP and CAP 10%: (a) CP – 0.5 Pa, (b) CP – 1320 Pa and (c) CAP 10% – 1320 Pa.

lets. All glasses exhibit this phenomenon at 120 °C whatever the relative humidity.

4. Discussion

4.1. Effect of Al_2O_3 content on physical parameters

Data in Table 2 clearly show that the addition of alumina lead in increase of T_g , Young's modulus and fracture toughness. This increase of physical properties is consistent with the alumina effect already encountered for sodium aluminophosphate glasses [7]. Metaphosphate glasses can be described as linear phosphate chains. Addition of small

alumina content strengthens the network by the cross-linking effect of Al–O₆ sites. Non-bridging PO[−]Ca²⁺ is replaced by covalent POAl bonds.

4.2. Stress corrosion model

Charles and Hillig and later Wiederhorn [2,14] have described stress corrosion domain by the following reaction-rate expression:

$$V_1 = A \left(\frac{P_{H_2O}}{P_0} \right)^m \exp \left(\frac{[K_I b - Q_1]}{RT} \right), \quad (2)$$

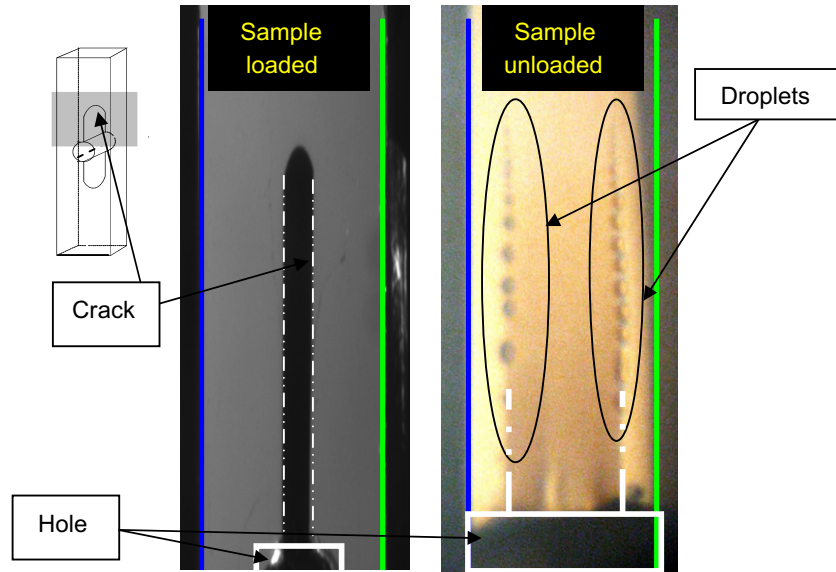


Fig. 6. Photo of the sample, just after unloading, when it is steel under experimental conditions. Dash lines are for the crack lines.

V_I (m/s) is the crack velocity in region I, P_{H_2O} (Pa) is the water vapor pressure, P_0 (Pa) is the atmospheric pressure, K_I (Pa.m^{1/2}) is the stress intensity factor, T (K) is the temperature and R is the gas constant. The model includes four parameters, experimentally determined – A , m , b and Q_I – that have various physical meanings in the reaction process: Q_I (J/mol) is the activation energy, m is the order of the reaction in terms of water vapor reactant, A (m/s) is a preexponential constant. b (m^{5/2}/mol) is related to the activation volume, V_a (m³/mol) and the crack tip radius, ρ (m), by the following expression:

$$b = \frac{2V_a}{3\sqrt{\pi\rho}}. \quad (3)$$

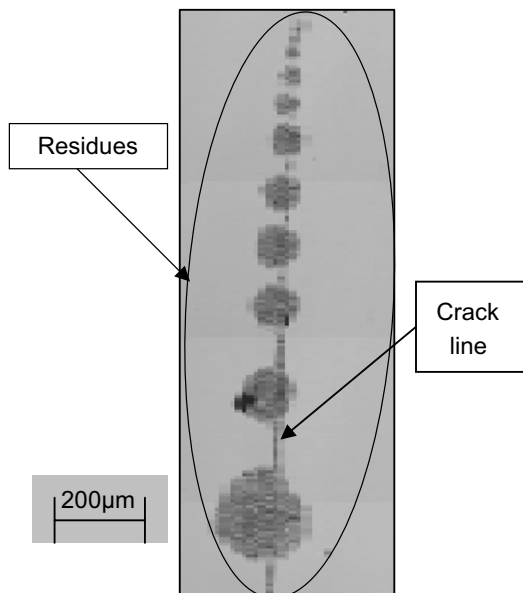


Fig. 7. Optical micrograph of crack line after holding the sample for a period of time of a few hours under ambient conditions.

Eq. (2) shows that under constant temperature and for a given stress intensity factor, an increase of the water vapor pressure induces an increase of the crack velocity. In the same way, under constant water vapor pressure and for a given stress intensity, a temperature increase induces an increase of the crack velocity. This model describes crack growth in many silicate and phosphate glasses and over a whole range of experimental conditions. Exceptions are encountered in regions where crack blunting or capillary condensation occurred. m parameter is generally 1 or 1.2 which indicates that the reaction is nearly first-order in H₂O.

Results at ambient temperature show an increase of the crack velocity with the water vapor pressure. However, Eq. (2) applied on our results (Fig. 3(b)) gives rise different m values. Between 0.5 Pa and 1320 Pa, m is 1.5 and between 1320 Pa and 2110 Pa m is 9. In order to quantify the gap between the model and our results, the dash line in Fig. 3(b) represents the predicted curve using Eq. (2) for 2110 Pa assuming $m = 1.2$. Activation energy, which is associated to a single mechanism, can not be calculated. However, from Eq. (2), we determine, for each composition, b and the activation volume, V_a , assuming a crack tip radius of 1.5 nm [15]. Values obtained are summarized in Table 3. They are similar to previously obtained values, for other glasses composition [15,16]. b is a parameter describing the stress fatigue. It is estimated from the slope

Table 3
 b and V_a of our phosphate glasses

Glasses	b (m ^{5/2} mol ^{−1})	V_a (10 ^{−6} m ³ mol ^{−1})
CP	0.28 ± 0.07	29 ± 7
CAP 2%	0.24 ± 0.06	25 ± 6
CAP 4%	0.33 ± 0.01	34 ± 1
CAP 10%	0.43 ± 0.02	44 ± 2

of $\log v_I - K_I$ curve. V_a , which is associated the volume under stresses at the crack tip, depends on the mean bond strength [17]. b increases with alumina content in the glass. This trend was already mentioned in static corrosion [8,9]. Then, V_a values show an improvement of the stress corrosion with the amount of alumina. Such evolution can be related to the increase of the network cross-linking and its strength as the Al_2O_3 content increase. These observations imply that the nature of the rate controlling reaction must be more complex than the one occurring in other phosphate glasses or silicate glasses.

4.3. Phenomenon associated to the droplets formation

From the observation at 120 °C, we assume the formation of a corrosion product on the fresh fracture surfaces which is expelled when the crack closes. This hypothesis is based on Watanabe et al. observations [17], who describe the formation of a ‘gel-like product’ on fresh fracture surfaces, for soda-lime glass according to different step: (a) water absorption onto the glass and alkali exchange in the near surface region, (b) formation of a ‘gel-like product’, and (c) nucleation and growth of a crystalline phase or phases. In case of metaphosphate glasses with CaO in their composition, $\text{Ca}^{2+}/\text{H}^+$ ion exchange reaction from the phosphate network inside the hydrated layer does not exist. This means that CP glass dissolve congruently by hydration reactions [18,19]. In phosphate glasses containing high Al_2O_3 content, dissolution mechanism is hydrolysis and a corrosion layer is observed [20]. Thus fresh fracture surface of glasses exhibit different corrosion mechanisms depending on their composition. CP glass reacts faster with water than do a glass containing alumina. Glasses containing Al_2O_3 develop a corrosion layer which modifies the stresses distribution at the crack tip. Effective K_I at the crack tip is probably different from calculated value.

We underline that our experiments extend over a few hours and $\log V - K_I$ curves are described from higher crack velocities to lowers ones. Behavior at 120 °C can be explained by changing stresses in alteration layers, or bridging between the crack surfaces. New experiments and chemical analyze of droplets residues are actually in progress in order to specify mechanisms which really take place.

5. Conclusion

Crack growth velocities are reported for a set of calcium aluminophosphate glasses over a range of temperature and water vapor pressure. Results are not in agreement with Wiederhorn’s model. Dependences on alumina content and temperature show unexpected behavior and a new phenomenon is observed at 120 °C when the sample is unloaded, droplets appear on the crack line. We assume that a corrosion mechanism probably superimposes on the stress corrosion phenomenon and affect the crack propagation.

Acknowledgments

P. Solignac for the operating system software, S. Clément, J.M. Fromental and J. Delmas for the experimental setup.

L. Duffours, Primeverre, for the preparation of the samples.

References

- [1] J. Charles, J. Appl. Phys. 29 (11) (1958).
- [2] S.M. Wiederhorn, J. Am. Ceram. Soc. 50 (8) (1967) 407.
- [3] S.N. Crichton et al., J. Am. Ceram. Soc. 82 (11) (1999) 3097.
- [4] Y.K. Lee, M. Tomozawa, J. Non-Cryst. Solids 248 (1999) 203.
- [5] T.I. Suratwala et al., J. Non-Cryst. Solids 263&264 (2000) 213.
- [6] T.I. Suratwala, R.A. Steele, J. Non-Cryst. Solids 316 (2003) 174.
- [7] A.E. Marino et al., J. Non-Cryst. Solids 289 (2001) 37.
- [8] R.K. Brow, J. Am. Ceram. Soc. 76 (4) (1993) 913.
- [9] D. Müller et al., Phys. Chem. Glasses 24 (2) (1983) 37.
- [10] C. Janssen, in: Proceeding of the 10th International Congress on Glass, vol. 10, Ceramic Society of Japan, Tokyo, Japan, 1974, p. 23.
- [11] M.Y. He, M.R. Turner, A.G. Evans, Acta Metall. Mater. 43 (1995) 3453.
- [12] E. Gehrke, Ch. Ullner, M. Hähner, J. Mater. Sci. 26 (1991) 5455.
- [13] T.A. Michalske, B.C. Bunker, K.D. Keefer, J. Non-Cryst. Solids 120 (1990) 126.
- [14] S.M. Wiederhorn et al., J. Am. Ceram. Soc. 57 (8) (1974) 336.
- [15] Y. Bando et al., J. Am. Ceram. Soc. 62 (5–6) (1984) C36.
- [16] S.M. Wiederhorn, L.H. Bolz, J. Am. Ceram. Soc. 53 (10) (1970) 543.
- [17] Y. Watanabe et al., J. Non-Cryst. Solids 177 (1994) 9.
- [18] B.C. Bunker, G.W. Arnold, J.A. Wilder, J. Non-Cryst. Solids 64 (1984) 291.
- [19] H. Gao, T. Tan, D. Wang, J. Non-Cryst. Solids 96 (2004) 29.
- [20] A. El Hadrami, M. Mesnaoui, M. Maazaz, J.-J. Videau, J. Non-Cryst. Solids 331 (2003) 228.